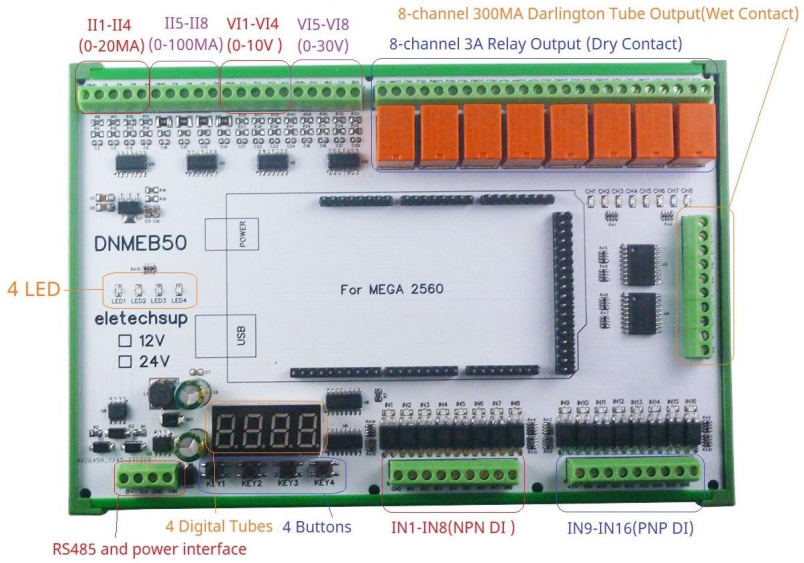
DNMEB50 MEGA2560 R3 Multifunctional expansion board



Product Features:

1 Operating voltage: DC 12V/DC 24V

2 Operating Current : Standby current (Digital tube OFF) 12MA, Standby current 20MA (Digital tube ON), 1 relay open 38MA... 8 relays open 150MA

3 Rich external expansion resources: 16ch NPN/PNP Digital Input;16ch Dry contact Wet contact digital output;16ch current and voltage analog input; RS485 Interface，4-digit digital tube, 4 LEDs, 4 buttons.

4 size 183*122*19mm(Only Board),188*128*29mm(with Din Rail Box)

5 Weight : 160g(Only Board), 293g(with Din Rail Box)

DNMEB50 Port	MEGA2560 R3 Pin
II1-II4(0-20MA)	A0-A3
II5-II8(0-100MA)	A4-A7
VI1-VI4(0-10V)	A8-A11
VI5-VI8(0-30V)	A12-A15
4 Digital Tubes	8/7/10/9
4 Buttons	21/20/12/11
4 LED	34/36/38/40
IN1-IN8(NPN DI)	6/5/4/3/2/14/15/19
IN9-IN16(PNP DI)	32/30/28/22/23/24/25/26
D1-D8(2803 DO)	41/39/37/35/33/31/29/27
CH1-CH8(Relay DO)	43/45/47/49/50/51/52/53

As long as you write ARDUINO code(sketch),You can use it to achieve a variety of delay timer function,such as:
RS485 Master-Slave Device(PLC MCU),
4x 0/4-20MA Current Collection,
4x 0-100MA Current Collection,
4x 0-5V/10V Voltage Collection,
4x 0-30V Voltage Collection,
Motor forward and reverse,
Timing on,
Timing off,
Power-up delay,
Trigger delay,
Infinite loop delay,
A finite number of cyclic delays,
Power sequencer,
And so on.

This is an expansion board based on MEGA2560 PRO, and the code(sketch) we provide is only used to test the experimental hardware.
If you need more functionality, write your own code. We do not provide additional code(sketch) and technical support

How to use: See video
" DNMEB50_RS485_Receiving_Sending_DEMO"
" DNMEB50_Analog_RS485_DEMO"
“DNMEB50_8NPN_8Relay_DEMO”

Note: DNMEB50 can not work independently, must be under the control of the MEGA2560 R3 board to work,If you do not have a MEGA2560 R3 board, purchase it separately or purchase a kit with MEGA2560 R3

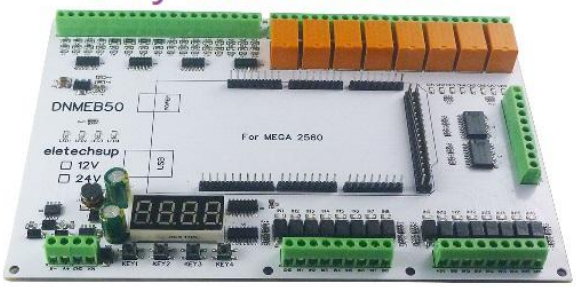
Insert MEGA2560 effect



With DIN Box version



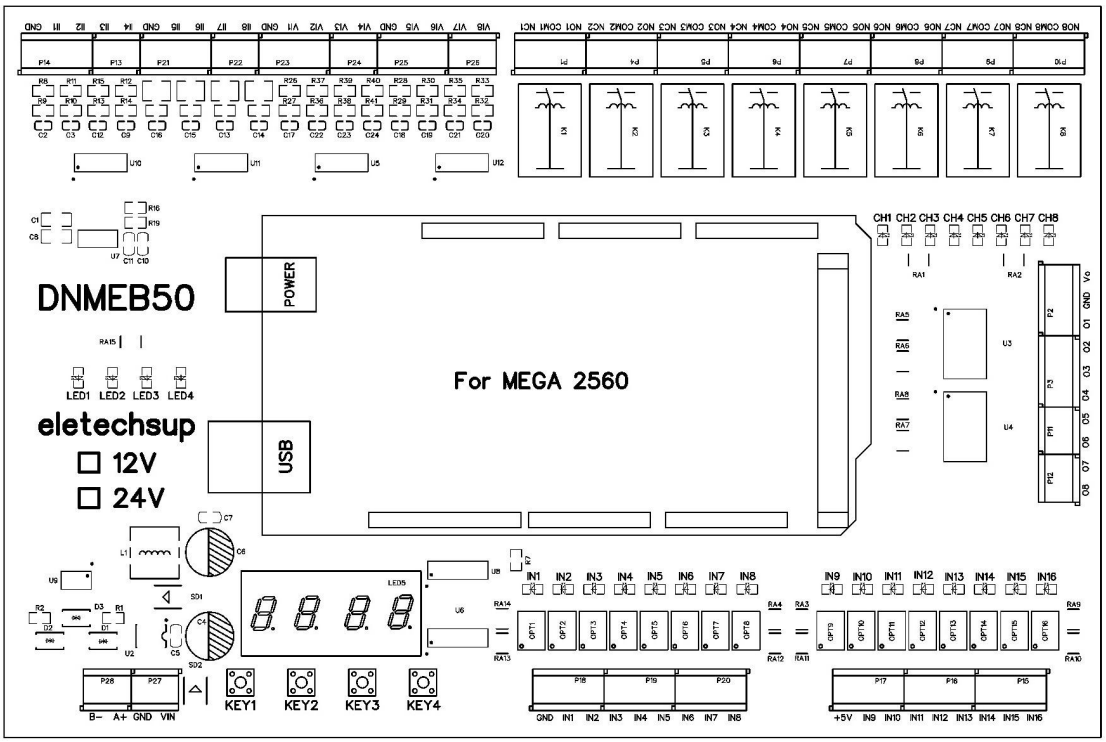
Only board version



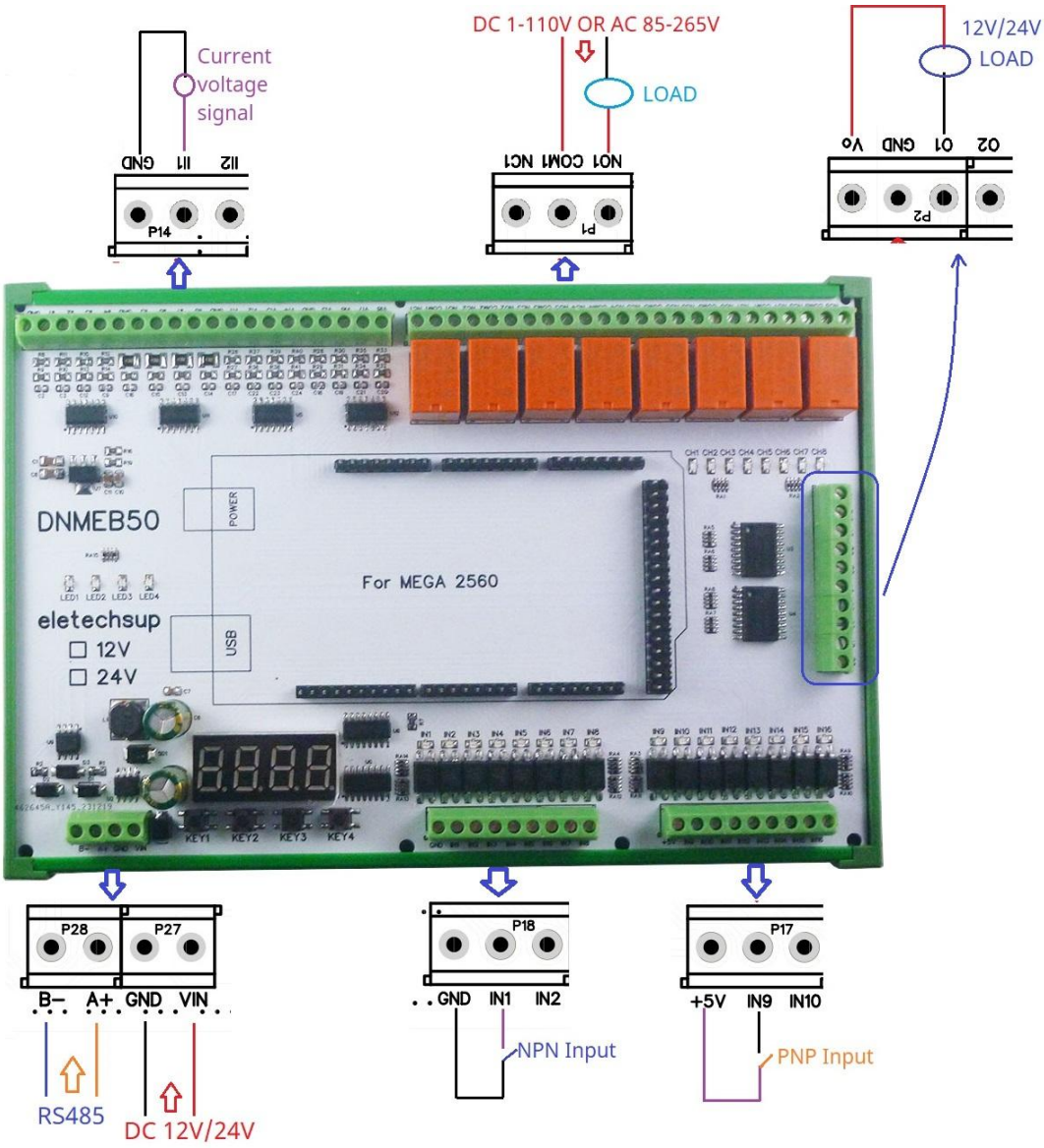
DIN Rail Box



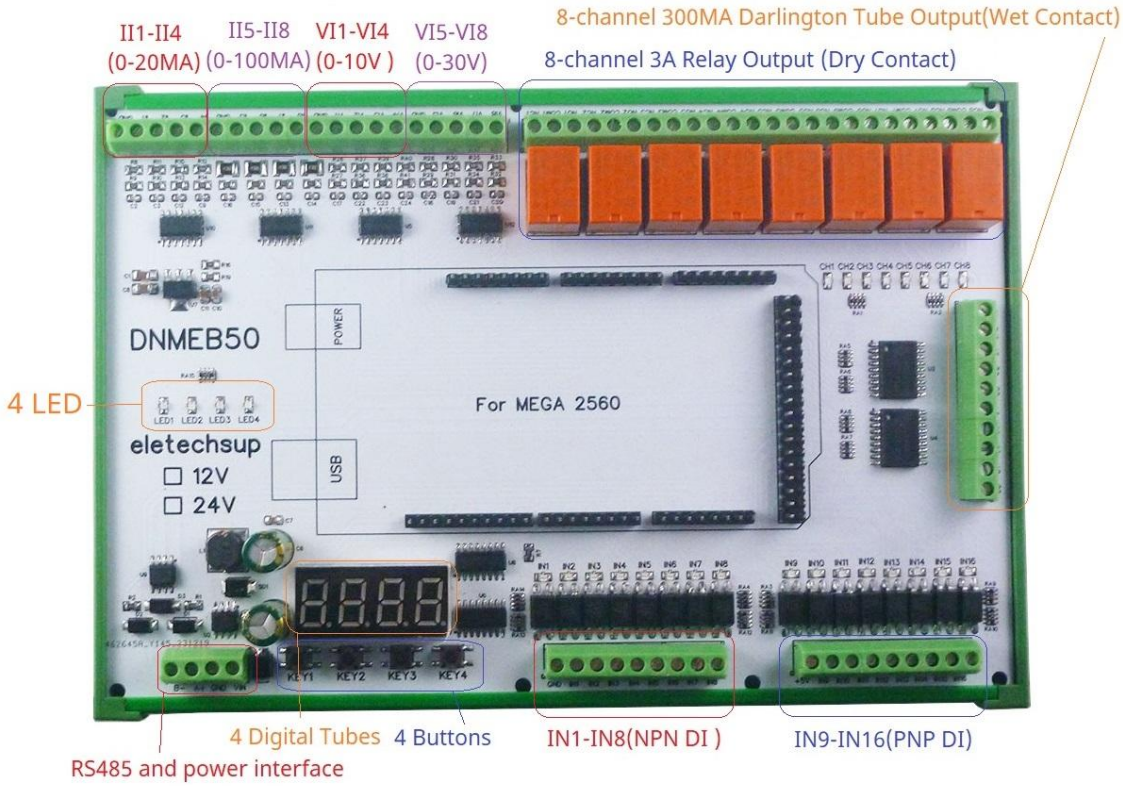
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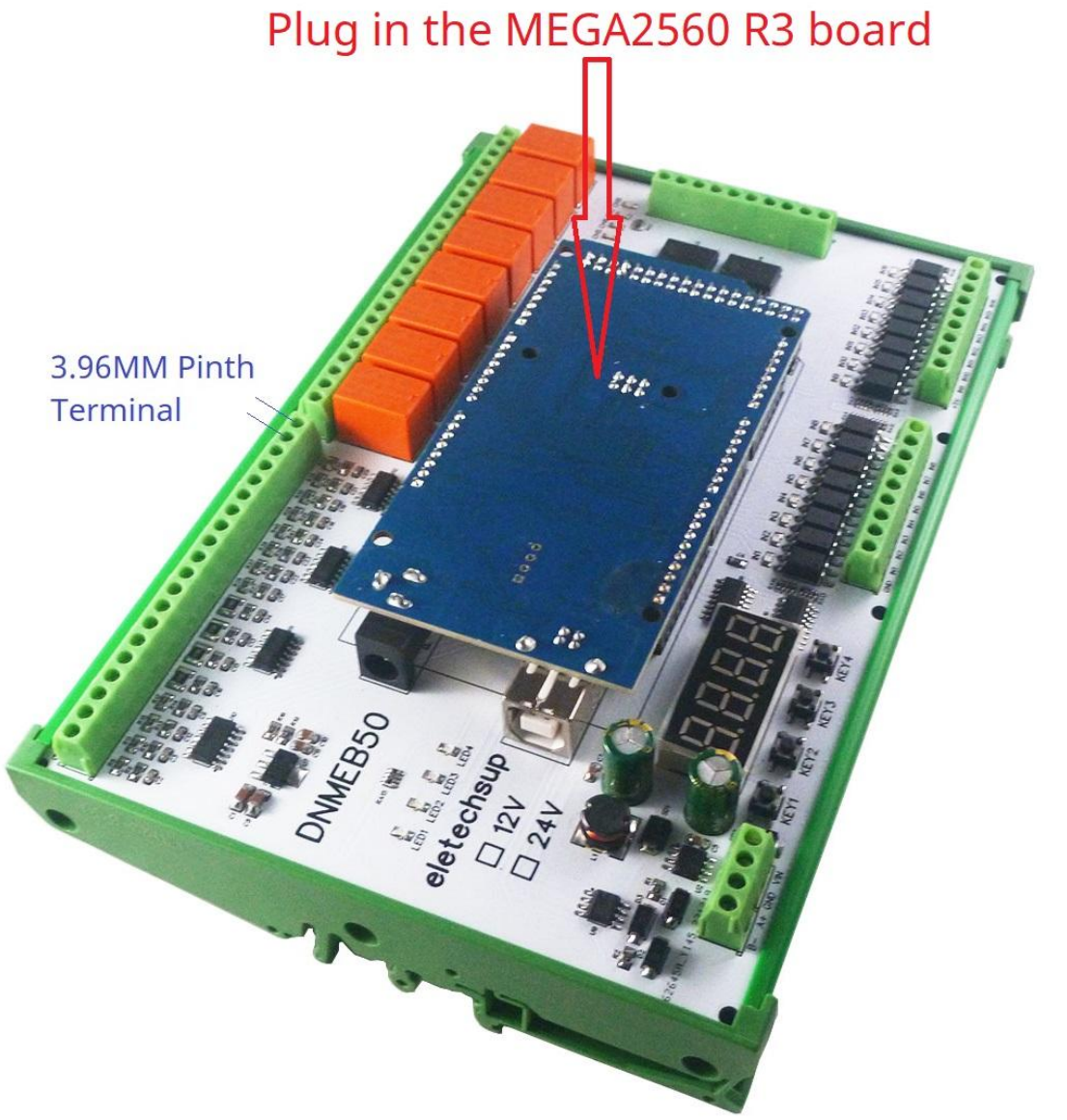

Current
voltage
signal
GND I1 I2
P14
DC 1-110V OR AC 85-265V
LOAD
NO1 COM1 NC1
P1
12V/24V
LOAD
GND O1 O2
P2
DNMEB50
POWER
For MEGA 2560
eletechsup
12V
24V
USB
KEY1 KEY2 KEY3 KEY4
P28 P27
B– A+ GND VIN
RS485
DC 12V/24V
P18
GND IN1 IN2
NPN Input
P17
+5V IN9 IN10
PNP Input

Onboard Resources



Port	MEGA Pin
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II5-II8(0-100MA)	A4-A7
VI1-VI4(0-10V)	A8-A11
VI5-VI8(0-30V)	A12-A15

Port	MEGA Pin
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IN9-IN16(PNP DI)	32/30/28/22/23/24/25/26
CH1-CH8(Relay DO)	41/39/37/35/33/31/29/27
D1-D8(2803 DO)	43/45/47/49/50/51/52/53



Without MEGA2560 R3, you need to bring your own